GOhack26: From Lagrange Interpolation to Threshold Cryptography for AIAgents

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Prover: _____

1. (1 point) An autonomous treasury agent can initiate and sign transactions using a private key stored on its server.

   An attacker compromises the agent. Which architectural change most directly limits the attacker's signing authority?

   A. Protect the private key using a hardware security module (HSM).

   B. Restrict the agent's permissions using access controls and transaction limits.

   C. Require independent threshold signers to authorize each transaction.

   D. Isolate the agent using a trusted execution environment (TEE).

2. (1 point) A 3-of-5 signing policy involves three humans, one policy engine, and one AI agent.

   What does this threshold guarantee?

   A. At least two humans must approve.

   B. At least one human must participate.

   C. The policy engine must approve.

   D. All three humans must approve.

3. (1 point) Two secret shares over the real numbers of a $(2, 3)$-scheme have been exfiltrated:

$$P_1 = (1, 19), \qquad P_3 = (3, 31).$$

   What is the value of the shared secret?

   A. 6

   B. 13

   C. 19

   D. 25

4. (1 point) Consider Shamir secret sharing over $\mathbb{F}_{17}$:

$$f(x) = 2 + 3x \pmod{17}.$$

Which share is invalid?

A. $(1, 5)$

B. $(7, 6)$

C. $(3, 10)$

D. $(11, 1)$

5. (1 point) A $(2, 3)$-Shamir scheme over $\mathbb{F}_{17}$ uses two shares:

$$P_1 = (1, 9), \qquad P_3 = (3, 15).$$

The Lagrange coefficients at $x = 0$ are $\lambda_1 = 10$ and $\lambda_3 = 8$.

What secret do these shares reconstruct?

A. 3

B. 6

C. 7

D. 12

6. (1 point) An attacker compromises three shares of a $(3, 4)$-Shamir secret sharing scheme over $\mathbb{F}_{23}$ and reconstructs the secret $s = 8$.

Which of the following polynomials could define a valid $(3, 4)$-Shamir scheme with this secret?

Select all that apply.

A. $f(x) = 8 + 2x$

B. $f(x) = 8 + 3x^2 + 2x$

C. $f(x) = 8 + 7x^2 + 14x$

D. $f(x) = 8 + x^3 + 17x^2 + x$

7. (1 point) Feldman's verifiable secret sharing (VSS) publishes commitments

$$C_0 = 7G, \qquad C_1 = 3G,$$

where G generates a cyclic group of order 17.

Participant $x = 1$ receives share $y = 11$.

Does the share satisfy

$$yG \stackrel{?}{=} C_0 + xC_1 \ ?$$

A. Yes, because $11 < 17$.

B. Yes, because the commitments are public.

C. No, because $11G \neq 10G$.

D. No, because three shares are required.

8. (1 point) In a toy distributed key generation (DKG) protocol over $\mathbb{F}_{17}$, three participants independently choose

$$f_1(x) = 3 + 2x,$$
$$f_2(x) = 5 + 4x,$$
$$f_3(x) = 2 + 7x.$$

What is the aggregated share at $x = 3$?

A. $(3, 6)$

B. $(3, 15)$

C. $(3, 10)$

D. $(3, 13)$

9. (1 point) In a simplified Schnorr signature example over a cyclic group of order 17, we have

$$PK = 7G, \qquad R = 10G, \qquad c = 2,$$

where G is the group generator, PK the public key, R the nonce commitment, and c the signature challenge.

Which response z satisfies

$$zG \stackrel{?}{=} R + cPK \ ?$$

A. 3

B. 7

C. 10

D. 14

10. (1 point) Which statement about FROST (Flexible Round-Optimized Schnorr Threshold Signatures) is correct?

A. The complete private key must be reconstructed before every signature.

B. Participants can jointly produce a Schnorr signature without reconstructing the private key during normal signing.

C. FROST signatures are post-quantum secure.

D. FROST guarantees that every signed transaction satisfies the organization's authorization policy.

11. (1 point) A team wants its autonomous-agent signing architecture to remain secure against future quantum attacks.

Select all that apply.

A. Shamir sharing retains information-theoretic secrecy when implemented correctly.

B. Feldman's discrete-logarithm commitments require a quantum-resistant replacement.

C. FROST becomes quantum-resistant automatically when its shares are generated by DKG.

D. Threshold ML-DSA is a research direction for post-quantum distributed signing.

12. (3 points) An autonomous agent proposes a malicious transaction. Three authorized threshold signers approve it, and the resulting signature verifies successfully.

    Did the cryptography fail? Justify your answer.